Observations and Tips Framework

Observations and Tips Framework

Purpose: Build a clear logic system for how Observations and Tips should be generated from session data in the Analysis section, based fully on your current approach to analysing professional cricketers.

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1. What an Insight Should Be

1.1 Observations

Observations should be:

- Pattern-based (repeated, not one-offs)
- Descriptive and objective
- Based on identifiable session behaviour
- Focused on what actually happened
- Free of coaching advice

Examples:

- Release height stayed stable across most deliveries
- Landing area concentrated in the back-of-length zone

1.2 Tips

Tips should be:

- Directly linked to a specific observation
- Actionable, short, and technically accurate
- Coach-like but simple
- Focused on improvement or reinforcement

Examples:

- Keep your wrist more stable at release to improve swing control
- Practise straight-bat drills for full-length balls

2. Data-to-Insight Workflow

You will define how raw session data becomes meaningful insights based on **how you currently analyse professional cricketers.**

2.1 Inputs to Use (Bowling + Batting)

All **batting and bowling parameters** should be defined by you exactly as they are used in professional cricket analysis.
This includes whatever you typically track today: release height, release angle, wrist position, line and length, timing quality, footwork patterns, head position, contact points, and any other parameters you consider essential at professional level.

You will specify:

- Which parameters matter for each skill
- How you interpret them in professional assessments
- How these parameters help identify patterns, strengths, or issues

2.2 Share Your Current Reporting Method

To align our app logic with real-world performance analysis, please share:

1. **Your current report format**
   The structure and style you use today when delivering analysis to players or coaches.
2. **A sample report (anonymised)**
   This helps us understand how you phrase insights, present findings, and structure conclusions.
3. **Your analysis process**
   Step-by-step breakdown of how you currently:
   - Observe a player
   - Identify patterns
   - Detect technical or tactical issues
   - Prioritise what matters most
   - Convert observations into coaching points
   - Decide what becomes a strength vs an area to improve
4. **How you define each key parameter**
   For example:
   - What counts as “good release height consistency”
   - What you consider “late contact” or “poor wrist position”
   - What qualifies as a biomechanical issue
   - How you label strengths vs problems

This ensures our system mirrors the way you already analyse professional cricketers.

3. Observations Framework

Define qualitative logic for each category below.

3.1 Consistency

Examples:

- Stable release height
- Stable bowling speed
- Consistent footwork pattern
- Repeatable shot selection for similar balls

3.2 Variation

Examples:

- Variation in release angle
- Timing inconsistency across shot types

3.3 Positional Patterns

Examples:

- Majority of balls landing short of length
- Playing cross-batted shots to fuller balls

3.4 Trend Patterns

Examples:

- Speed increasing or dropping over time
- Timing improving across the session

3.5 Biomechanical Stability

Examples:

- Wrist stable at release
- Head steady during shots
- Shoulder alignment consistent

3.6 Strengths

Examples:

- Strong timing on back-foot shots
- High consistency in off-stump targeting

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4. Tips Framework

Every Tip must map directly to an Observation.

4.1 Tip Types

- Technical
- Tactical
- Consistency
- Physical / load indicators

4.2 Tip Rules

- Only generate tips when patterns repeat clearly
- Avoid tips based on very small sample size
- Avoid showing too many tips
- Each tip addresses one behaviour
- Tips should be practical and simple

4.3 Tip Template

1. What to improve or reinforce
2. Why it matters
3. What to work on next session

4.4 Tip Style

- Short
- Action-oriented
- Player-friendly
- Optional drill or practice suggestion

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5. Observation → Tip Mapping

You will build a full mapping table.

Example Mapping

Observation	Interpretation	Tip
Variation in release height	Wrist inconsistency or timing issue	Work on wrist stability at release
Majority short-of-length	Release timing or load-up delay	Practise driving front foot into the crease
Late timing on back-foot shots	Weight transfer or head drop	Work on earlier weight transfer
Speed drops toward the end	Fatigue or rhythm loss	Focus on consistent run-up rhythm

Role-Based Extensions

Fast Bowling

Observation	Interpretation	Tip	Drill
Wrist angle variation at release	Late pronation timing	Focus on wrist snap timing	One-knee wrist snap drill

Spin Bowling

Observation	Interpretation	Tip	Drill
Drop in rev rate	Shoulder rotation delay	Lead with elbow higher	Wall pivot rotation drill

Batting

Observation	Interpretation	Tip	Drill
Late contact vs bouncers	Late transfer / head drop	Maintain head forward longer	High-bounce machine drill

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6. Decision Rules

6.1 When to Trigger an Observation

- Pattern repeats meaningfully
- Clear clustering
- Trend persists
- Enough data
- Not caused by detection noise

6.2 When NOT to Trigger

- Too few balls
- Poor camera angle
- One-off abnormal deliveries
- Low confidence

6.3 When to Trigger a Tip

- Observation repeats enough to matter
- Insight is actionable
- Not redundant or repeated too often
- Data is sufficient

6.4 Prioritisation Rules

Prioritise insights that are:

- Most important
- Most repeated
- Most impactful
- Most relevant to the player's role

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7. Deliverables You Must Produce

7.1 Observations Library

40–60 observations covering batting and bowling.
Each includes:

- Meaning
- Trigger pattern (qualitative)
- Example wording
- Contextual points as per eg fast bowler so on

7.2 Tips Library

40–60 tips.
Each includes:

- Tip wording
- Mapped observation
- Skill category
- Optional drill

7.3 Insight Logic Document

Explaining:

- How each parameter is used
- What counts as a pattern
- Confidence rules
- Priority rules
- Suppression rules

7.4 Data Interpretation Guide

Detail:

- How to interpret line, length, timing, mechanics, speed
- Typical behavioural signatures
- How to avoid misinterpreting patterns

7.5 Review & Iteration Plan

Explain:

- How insights will be validated
- How thresholds will evolve after real user data
- How coach feedback will be incorporated

Current Analysis Framework

Analytical Engine

Bowler Performance Analysis

The analytical engine evaluates bowling performance using ball-tracking data (generated via computer vision) and biomechanical cues from video.

Data Sources

- Computer Vision Data: Quantitative metrics including:
    - Ball speed (km/h) per delivery
    - Ball length (distance from stumps in meters)
    - Key event frames (release, bounce, bat contact)
    - Trajectory tracking
- Video Analysis: Frame-by-frame video reasoning to assess biomechanical patterns (dev in progress)

What the Engine Detects (dev in progress)

- Run-up rhythm
- Jump phase and gather position stability
- Front-foot landing mechanics
- Release point
- Repeatability across deliveries
- Speed profile
- Trajectory shape and bounce position
- Biomechanical

What the Engine Outputs

- Objective observations
- Tips linked to actionable corrections targeting performance improvement

Batsman Performance Analysis

The engine evaluates batting technique and response to delivery types using video reasoning.

Data Sources

Video Analysis: Frame-by-frame video reasoning to assess technique

What the Engine Detects

- Batting stance and setup (head stability, grip, alignment, balance)
- Footwork (movement to pitch of the ball, weight transfer, stability)
- Shot execution (timing, contact quality, bat path, follow-through)
- Response to different ball lengths and lines

What the Engine Outputs

- Objective observations
- Tips linked to actionable corrections targeting performance improvement

Future Work

- **Pose Estimation Integration** – Implement pose estimation to measure body alignment, joint angles, and movement patterns for both bowling and batting actions, enabling precise biomechanical assessment across all key phases (run-up to follow-through for bowlers, stance to follow-through for batsmen).

- **Individual Player Tagging & Tracking** – Introduce automated player identification within multi-player training videos to allow per-player performance tracking, comparison, and long-term development monitoring.

- **Knowledge-Supported Coaching Insights** – Link analytical engine outputs with a cricket knowledge base of expert coaching references, drills, and technique guides, allowing the system to enhance observations and tips with validated context and recommendations.

Sample Sessions

In **Session 1**, the AI engine originally produced results in the **strengths & weaknesses format**. We have now moved to the new **observations & tips structure**. The same Session 1 video was reprocessed using the new format to enable comparison.

In **Session 2** (latest recording), the analysis was generated directly in the observations & tips format.

Ball Tracking Data Input to the Analysis Engine

Before generating observations and tips, the model receives structured computer-vision data extracted from the session video. For each delivery, the system captures raw measurements such as:

- Ball trajectory through the air (frame-by-frame 2D/3D tracking)
- Release point (position and height at release)
- Bounce point and delivery length relative to stumps
- Ball speed at release
- Pitch map placement
- Bat contact events

Session 1

Video - 008330cf-9d1e-41b5-8bcf-7a233788ad3a.mp4

Initial Output (Strengths & Weaknesses Format)


```
"analysis_report": {
 "analysis": {
   "strengths": [
     "Generates good pace with an energetic and committed action.",
     "Consistently bowls an attacking, full length aimed at the stumps.",
     "Maintains a good wrist position at release, promoting a straight seam.",
     "Shows good intent and commitment in the delivery stride."
   ],
   "weakness": [
     "Inconsistent line and control, often falling away in the follow-through.",
     "Front knee collapses upon landing, leading to a loss of momentum and stability.",
     "Mixed bowling action (neither fully side-on nor front-on) increases the risk of back and shoulder injuries.",
     "Lacks variation in length, tending to bowl predictably full."
   ]
 }
}
```

New Output Format After Model Enhancement

```
"analysis_report": {
  "bowler": {
    "observations": [
      "The run-up lacks consistent rhythm, affecting momentum into the delivery stride.",
      "Front leg collapses slightly upon landing, reducing stability and energy transfer.",
      "Follow-through momentum frequently falls away towards the off-side after release.",
      "A high arm path creates a good foundation for generating bounce.",
      "Bounce point data indicates significant inconsistency in line and length control."
    ],
    "tips": [
      "Develop a smoother, accelerating run-up to build consistent delivery momentum.",
      "Focus on landing with a firm, braced front leg for a stable base.",
      "Drive your body's momentum directly towards the target after releasing the ball.",
      "Maintain a stable head position throughout the action to improve control.",
      "Use the non-bowling arm to pull down forcefully, keeping the action aligned."
    ]
  }
},
"batsmen": {
  "observations": [
    "The head remains commendably still through the point of contact.",
    "A short front-foot stride leads to reaching for fuller deliveries.",
    "The batsman frequently plays from the crease with minimal forward movement.",
    "Drives are often played aerially due to an early bat-face opening.",
    "Follow-through is consistently abbreviated, limiting full shot completion."
  ],
  "tips": [
    "Practice a longer, more committed stride towards the pitch of the ball.",
    "Develop a small forward press to initiate positive weight transfer.",
    "Focus on keeping your front shoulder and elbow high during drives.",
    "Meet the ball closer to your body and directly under your eyes.",
    "Allow the bat to swing completely through the line of the shot."
  ]
}}
```

Session 2

Video - thursday_afternoon_test_session_.mp4


```
"analysis_report": {
  "bowler": {
    "observations": [
      "Release height is maintained with excellent consistency across deliveries.",
      "There is significant, unintentional variation in bowling speed.",
      "Control over line and length is inconsistent between balls.",
      "The follow-through consistently falls away towards the off-side.",
      "A short run-up limits momentum generation before the delivery stride."
    ],
    "tips": [
      "Develop a longer, more rhythmic run-up to build consistent momentum.",
      "Focus on driving your body's momentum directly towards the target.",
      "Strengthen your front leg brace to provide a stable base for rotation.",
      "Practice bowling at a set pace to improve arm speed consistency.",
      "Use target-based drills to improve accuracy of line and length."
    ]
  }
},
"batsmen": {
  "observations": [
    "Head falls outside the line of the front foot during forward shots.",
    "Front foot stride is short, causing the batsman to reach for the ball.",
    "Bat path comes around the front pad instead of straight down the line.",
    "Incomplete weight transfer leads to off-balance shots and reduced power.",
    "Shoulders are slightly open in the stance, promoting an across-the-line swing."
  ],
  "tips": [
    "Focus on keeping your head over the ball during the front-foot drive.",
    "Lengthen your front-foot stride to get closer to the pitch of the ball.",
    "Practice hitting down a straight line to ensure a straighter bat path.",
    "Commit your body weight fully into the shot over a stable front leg.",
    "Align your front shoulder more towards the bowler for better side-on positioning."
  ]
}}
```